Red Hat OpenShift Data Foundation 4.22

Replacing devices

Instructions for safely replacing operational or failed devices

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Instructions for safely replacing operational or failed devices

Abstract

This document explains how to safely replace storage devices for Red Hat OpenShift Data Foundation.

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PREFACE

Depending on the type of your deployment, you can choose one of the following procedures to replace a storage device:

- For dynamically created storage clusters deployed on AWS, see:
    - Section 1.1, "Replacing operational or failed storage devices on AWS user-provisioned infrastructure".
    - Section 1.2, "Replacing operational or failed storage devices on AWS installer-provisioned infrastructure".
- For dynamically created storage clusters deployed on VMware, see Section 2.1, "Replacing operational or failed storage devices on VMware infrastructure".
- For dynamically created storage clusters deployed on Microsoft Azure, see Section 3.1, "Replacing operational or failed storage devices on Azure installer-provisioned infrastructure".
- For storage clusters deployed using local storage devices, see:
    - Section 5.1, "Replacing operational or failed storage devices on clusters backed by local storage devices".
    - Section 5.2, "Replacing operational or failed storage devices on IBM Power" .

NOTE

OpenShift Data Foundation does not support heterogeneous OSD sizes.

CHAPTER 1. DYNAMICALLY PROVISIONED OPENSHIFT DATA FOUNDATION DEPLOYED ON AWS

1.1. REPLACING OPERATIONAL OR FAILED STORAGE DEVICES ON AWS USER-PROVISIONED INFRASTRUCTURE

When you need to replace a device in a dynamically created storage cluster on an AWS user-provisioned infrastructure, you must replace the storage node. For information about how to replace nodes, see:

- Replacing an operational AWS node on user-provisioned infrastructure .
- Replacing a failed AWS node on user-provisioned infrastructure .

1.2. REPLACING OPERATIONAL OR FAILED STORAGE DEVICES ON AWS INSTALLER-PROVISIONED INFRASTRUCTURE

When you need to replace a device in a dynamically created storage cluster on an AWS installer-provisioned infrastructure, you must replace the storage node. For information about how to replace nodes, see:

- Replacing an operational AWS node on installer-provisioned infrastructure .
- Replacing a failed AWS node on installer-provisioned infrastructure .

CHAPTER 2. DYNAMICALLY PROVISIONED OPENSHIFT DATA FOUNDATION DEPLOYED ON VMWARE

2.1. REPLACING OPERATIONAL OR FAILED STORAGE DEVICES ON VMWARE INFRASTRUCTURE

Create a new Persistent Volume Claim (PVC) on a new volume, and remove the old object storage device (OSD) when one or more virtual machine disks (VMDK) needs to be replaced in OpenShift Data Foundation which is deployed dynamically on VMware infrastructure.

Prerequisites

- Ensure that the data is resilient.
  - In the OpenShift Web Console, click **Storage → Data Foundation**.
  - Click the **Storage Systems** tab, and then click **ocs-storagecluster**.
  - In the **Status card** of **Block and File** dashboard, under the **Overview** tab, verify that *Data Resiliency* has a green tick mark.

Procedure

1. Identify the OSD that needs to be replaced and the OpenShift Container Platform node that has the OSD scheduled on it.

   ```
   $ oc get -n openshift-storage pods -l app=rook-ceph-osd -o wide
   ```

   Example output:

   ```
   rook-ceph-osd-0-6d77d6c7c6-m8xj6    0/1    CrashLoopBackOff    0    24h   10.129.0.16   compute-2   <none>           <none>
   rook-ceph-osd-1-85d99fb95f-2svc7    1/1    Running             0    24h   10.128.2.24   compute-0   <none>           <none>
   rook-ceph-osd-2-6c66cdb977-jp542    1/1    Running             0    24h   10.130.0.18   compute-1   <none>           <none>
   ```

   In this example, **rook-ceph-osd-0-6d77d6c7c6-m8xj6** needs to be replaced and **compute-2** is the OpenShift Container platform node on which the OSD is scheduled.

   NOTE

   The status of the pod is **Running**, if the OSD you want to replace is healthy.

2. Scale down the OSD deployment for the OSD to be replaced.
   Each time you want to replace the OSD, update the **osd_id_to_remove** parameter with the OSD ID, and repeat this step.

   ```
   $ osd_id_to_remove=0
   ```

   ```
   $ oc scale -n openshift-storage deployment rook-ceph-osd-${osd_id_to_remove} --replicas=0
   ```

where, **osd_id_to_remove** is the integer in the pod name immediately after the **rook-ceph-osd** prefix. In this example, the deployment name is **rook-ceph-osd-0**.

Example output:

```
deployment.extensions/rook-ceph-osd-0 scaled
```

3. Verify that the **rook-ceph-osd** pod is terminated.

```
$ oc get -n openshift-storage pods -l ceph-osd-id=${osd_id_to_remove}
```

Example output:

```
No resources found.
```

IMPORTANT

If the **rook-ceph-osd** pod is in **terminating** state, use the **force** option to delete the pod.

```
$ oc delete pod rook-ceph-osd-0-6d77d6c7c6-m8xj6 --force --grace-period=0
```

Example output:

```
warning: Immediate deletion does not wait for confirmation that the running
resource has been terminated. The resource may continue to run on the
cluster indefinitely.
  pod "rook-ceph-osd-0-6d77d6c7c6-m8xj6" force deleted
```

4. Remove the old OSD from the cluster so that you can add a new OSD.

   a. Delete any old **ocs-osd-removal** jobs.

   ```
   $ oc delete -n openshift-storage job ocs-osd-removal-job
   ```

   Example output:

   ```
   job.batch "ocs-osd-removal-job" deleted
   ```

   NOTE

   If the above job does not reach **Completed** state after 10 minutes, then the job must be deleted and rerun with **FORCE_OSD_REMOVAL=true**.

   b. Navigate to the **openshift-storage** project.

   ```
   $ oc project openshift-storage
   ```

   c. Remove the old OSD from the cluster.

```
$ oc process -n openshift-storage ocs-osd-removal -p
FAILED_OSD_IDS=${osd_id_to_remove} FORCE_OSD_REMOVAL=false |oc create -n
openshift-storage -f -
```

The FORCE_OSD_REMOVAL value must be changed to “true” in clusters that only have three OSDs, or clusters with insufficient space to restore all three replicas of the data after the OSD is removed.

> **WARNING**
>
> Proceeding will permanently remove the OSD from the cluster and destroy its data. Ensure **osd_id_to_remove** is correct. Do not process multiple IDs from the same failure domain at once. Contact Red Hat Technical Support if you need assistance.

5. Verify that the OSD was removed successfully by checking the status of the **ocs-osd-removal-job** pod.
   A status of **Completed** confirms that the OSD removal job succeeded.

   ```
   $ oc get pod -l job-name=ocs-osd-removal-job -n openshift-storage
   ```

6. Ensure that the OSD removal is completed.

   ```
   $ oc logs -l job-name=ocs-osd-removal-job -n openshift-storage --tail=-1 | egrep -i 'completed
   removal'
   ```

   Example output:

   ```
   2022-05-10 06:50:04.501511 I | cephosd: completed removal of OSD 0
   ```

   > **IMPORTANT**
   >
   > If the **ocs-osd-removal-job** pod fails and the pod is not in the expected **Completed** state, check the pod logs for further debugging.
   >
   > For example:
   >
   > ```
   > # oc logs -l job-name=ocs-osd-removal-job -n openshift-storage --tail=-1
   > ```

7. If encryption was enabled at the time of install, remove **dm-crypt** managed **device-mapper** mapping from the OSD devices that are removed from the respective OpenShift Data Foundation nodes.

   a. Get the PVC name(s) of the replaced OSD(s) from the logs of **ocs-osd-removal-job** pod.

      ```
      $ oc logs -l job-name=ocs-osd-removal-job -n openshift-storage --tail=-1  |egrep -i
      ‘pvc|deviceset’
      ```

Example output:

```
2021-05-12 14:31:34.666000 I | cephosd: removing the OSD PVC "ocs-deviceset-xxxx-xxx-xxx-xxx"
```

b. For each of the previously identified nodes, do the following:

   i. Create a **debug** pod and **chroot** to the host on the storage node.

   ```
   $ oc debug node/<node name>
   ```

   <node name>

   Is the name of the node.

   ```
   $ chroot /host
   ```

   ii. Find a relevant device name based on the PVC names identified in the previous step.

   ```
   $ dmsetup ls| grep <pvc name>
   ```

   <pvc name>

   Is the name of the PVC.
   Example output:

   ```
   ocs-deviceset-xxx-xxx-xxx-xxx-block-dmcrypt (253:0)
   ```

   iii. Remove the mapped device.

   ```
   $ cryptsetup luksClose --debug --verbose ocs-deviceset-xxx-xxx-xxx-xxx-block-dmcrypt
   ```

IMPORTANT

If the above command gets stuck due to insufficient privileges, run the following commands:

- Press **CTRL+Z** to exit the above command.
- Find the PID of the process which was stuck.

```
$ ps -ef | grep crypt
```

- Terminate the process using the **kill** command.

```
$ kill -9 <PID>
```

<PID>

Is the process ID.

- Verify that the device name is removed.

```
$ dmsetup ls
```

8. Delete the **ocs-osd-removal** job.

```
$ oc delete -n openshift-storage job ocs-osd-removal-job
```

Example output:

```
job.batch "ocs-osd-removal-job" deleted
```

NOTE

When using an external key management system (KMS) with data encryption, the old OSD encryption key can be removed from the Vault server as it is now an orphan key.

Verification steps

1. Verify that there is a new OSD running.

```
$ oc get -n openshift-storage pods -l app=rook-ceph-osd
```

Example output:

```
rook-ceph-osd-0-5f7f4747d4-snshw                    1/1     Running     0          4m47s
rook-ceph-osd-1-85d99fb95f-2svc7                    1/1     Running     0          1d20h
rook-ceph-osd-2-6c66cdb977-jp542                    1/1     Running     0          1d20h
```

2. Verify that there is a new PVC created which is in **Bound** state.

```
$ oc get -n openshift-storage pvc
```

Example output:

```
NAME                 STATUS   VOLUME                                     CAPACITY   ACCESS
MODES   STORAGECLASS   AGE
ocs-deviceset-0-0-2s6w4   Bound    pvc-7c9bcaf7-de68-40e1-95f9-0b0d7c0ae2fc   512Gi
RWO          thin           5m
ocs-deviceset-1-0-q8fwh   Bound    pvc-9e7e00cb-6b33-402e-9dc5-b8df4fd9010f   512Gi
RWO          thin           1d20h
ocs-deviceset-2-0-9v8lq   Bound    pvc-38cdfcee-ea7e-42a5-a6e1-aaa6d4924291   512Gi
RWO          thin           1d20h
```

3. Optional: If cluster-wide encryption is enabled on the cluster, verify that the new OSD devices are encrypted.

   a. Identify the nodes where the new OSD pods are running.

      ```
      $ oc get -n openshift-storage -o=custom-columns=NODE:.spec.nodeName pod/<OSD-pod-name>
      ```

      <OSD-pod-name>

      Is the name of the OSD pod.
      For example:

      ```
      $ oc get -n openshift-storage -o=custom-columns=NODE:.spec.nodeName pod/rook-ceph-osd-0-544db49d7f-qrgqm
      ```

      Example output:

      ```
      NODE
      compute-1
      ```

   b. For each of the nodes identified in the previous step, do the following:

      i. Create a debug pod and open a chroot environment for the selected host(s).

         ```
         $ oc debug node/<node name>
         ```

         <node name>

         Is the name of the node.

         ```
         $ chroot /host
         ```

      ii. Check for the **crypt** keyword beside the **ocs-deviceset** name(s).

         ```
         $ lsblk
         ```

4. Log in to OpenShift Web Console and view the storage dashboard.

CHAPTER 3. DYNAMICALLY PROVISIONED OPENSHIFT DATA FOUNDATION DEPLOYED ON MICROSOFT AZURE

3.1. REPLACING OPERATIONAL OR FAILED STORAGE DEVICES ON AZURE INSTALLER-PROVISIONED INFRASTRUCTURE

When you need to replace a device in a dynamically created storage cluster on an Azure installer-provisioned infrastructure, you must replace the storage node. For information about how to replace nodes, see:

- Replacing operational nodes on Azure installer-provisioned infrastructure.
- Replacing failed nodes on Azure installer-provisioned infrastructures.

CHAPTER 4. DYNAMICALLY PROVISIONED OPENSHIFT DATA FOUNDATION DEPLOYED ON GOOGLE CLOUD

4.1. REPLACING OPERATIONAL OR FAILED STORAGE DEVICES ON GOOGLE CLOUD INSTALLER-PROVISIONED INFRASTRUCTURE

When you need to replace a device in a dynamically created storage cluster on an Google Cloud installer-provisioned infrastructure, you must replace the storage node. For information about how to replace nodes, see:

- Replacing operational nodes on Google Cloud installer-provisioned infrastructure
- Replacing failed nodes on Google Cloud installer-provisioned infrastructures .

CHAPTER 5. OPENSHIFT DATA FOUNDATION DEPLOYED USING LOCAL STORAGE DEVICES

5.1. REPLACING OPERATIONAL OR FAILED STORAGE DEVICES ON CLUSTERS BACKED BY LOCAL STORAGE DEVICES

You can replace an object storage device (OSD) in OpenShift Data Foundation deployed using local storage devices on the following infrastructures:

- Bare metal / Agnostic deployment
- VMware
- IBM Z and IBM® LinuxONE

> **NOTE**
>
> There might be a need to replace one or more underlying storage devices.

Prerequisites

- Red Hat recommends that replacement devices are configured with similar infrastructure and resources to the device being replaced.
- Ensure that the data is resilient.
  - In the OpenShift Web Console, click **Storage → Data Foundation**.
  - Click the **Storage Systems** tab, and then click **ocs-storagecluster**.
  - In the **Status card** of **Block and File** dashboard, under the **Overview** tab, verify that *Data Resiliency* has a green tick mark.

Procedure

1. Remove the underlying storage device from relevant worker node.
2. Verify that relevant OSD Pod has moved to CrashLoopBackOff state.
   Identify the OSD that needs to be replaced and the OpenShift Container Platform node that has the OSD scheduled on it.

```
$ oc get -n openshift-storage pods -l app=rook-ceph-osd -o wide
```

   Example output:

```
rook-ceph-osd-0-6d77d6c7c6-m8xj6    0/1    CrashLoopBackOff    0    24h    10.129.0.16
compute-2   <none>           <none>
rook-ceph-osd-1-85d99fb95f-2svc7    1/1    Running           0    24h    10.128.2.24   compute-
0   <none>           <none>
rook-ceph-osd-2-6c66cdb977-jp542    1/1    Running           0    24h    10.130.0.18   compute-
1   <none>           <none>
```

In this example, **rook-ceph-osd-0-6d77d6c7c6-m8xj6** needs to be replaced and **compute-2** is the OpenShift Container platform node on which the OSD is scheduled.

3. Scale down the **rook-ceph-operator** deployment.

```
$ oc -n openshift-storage scale deployment rook-ceph-operator --replicas=0
```

4. Scale down the OSD deployment for the OSD to be replaced.

```
$ osd_id_to_remove=0
```

```
$ oc scale -n openshift-storage deployment rook-ceph-osd-${osd_id_to_remove} --replicas=0
```

where, **osd_id_to_remove** is the integer in the pod name immediately after the **rook-ceph-osd** prefix. In this example, the deployment name is **rook-ceph-osd-0**.

Example output:

```
deployment.extensions/rook-ceph-osd-0 scaled
```

5. Verify that the **rook-ceph-osd** pod is terminated.

```
$ oc get -n openshift-storage pods -l ceph-osd-id=${osd_id_to_remove}
```

Example output:

```
No resources found in openshift-storage namespace.
```

> **IMPORTANT**
>
> If the **rook-ceph-osd** pod is in **terminating** state for more than a few minutes, use the **force** option to delete the pod.
>
> ```
> $ oc delete -n openshift-storage pod rook-ceph-osd-0-6d77d6c7c6-m8xj6 --grace-period=0 --force
> ```
>
> Example output:
>
> ```
> warning: Immediate deletion does not wait for confirmation that the running resource has been terminated. The resource may continue to run on the cluster indefinitely.
>   pod "rook-ceph-osd-0-6d77d6c7c6-m8xj6" force deleted
> ```

6. Remove the old OSD from the cluster so that you can add a new OSD.

   a. Delete any old **ocs-osd-removal** jobs.

   ```
   $ oc delete -n openshift-storage job ocs-osd-removal-job
   ```

   Example output:

```
job.batch "ocs-osd-removal-job" deleted
```

NOTE

The above command must reach **Completed** state before moving to the next steps. This can take more than ten minutes.

b. Navigate to the **openshift-storage** project.

```
$ oc project openshift-storage
```

c. Remove the old OSD from the cluster.

```
$ oc process -n openshift-storage ocs-osd-removal -p
FAILED_OSD_IDS=${osd_id_to_remove} FORCE_OSD_REMOVAL=false |oc create -n
openshift-storage -f -
```

The FORCE_OSD_REMOVAL value must be changed to “true” in clusters that only have three OSDs, or clusters with insufficient space to restore all three replicas of the data after the OSD is removed.

WARNING

Proceeding will permanently remove the OSD from the cluster and destroy its data. Ensure **osd_id_to_remove** is correct. Do not process multiple IDs from the same failure domain at once. Contact Red Hat Technical Support if you need assistance.

7. Verify that the OSD was removed successfully by checking the status of the **ocs-osd-removal-job** pod.
A status of **Completed** confirms that the OSD removal job succeeded.

```
$ oc get pod -l job-name=ocs-osd-removal-job -n openshift-storage
```

8. Ensure that the OSD removal is completed.

```
$ oc logs -l job-name=ocs-osd-removal-job -n openshift-storage --tail=-1 | egrep -i 'completed
removal'
```

Example output:

```
2022-05-10 06:50:04.501511 I | cephosd: completed removal of OSD 0
```

> **IMPORTANT**
>
> If the **ocs-osd-removal-job** fails and the pod is not in the expected **Completed** state, check the pod logs for further debugging.
>
> For example:
>
> ```
> # oc logs -l job-name=ocs-osd-removal-job -n openshift-storage --tail=-1
> ```

9. If encryption was enabled at the time of install, remove **dm-crypt** managed **device-mapper** mapping from the OSDs that are removed from the respective OpenShift Data Foundation nodes.

    a. Get the Persistent Volume Claim (PVC) names of the replaced OSDs from the logs of **ocs-osd-removal-job** pod.

    ```
    $ oc logs -l job-name=ocs-osd-removal-job -n openshift-storage --tail=-1  |egrep -i
    'pvc|deviceset'
    ```

    Example output:

    ```
    2021-05-12 14:31:34.666000 I | cephosd: removing the OSD PVC "ocs-deviceset-xxxx-
    xxx-xxx-xxx"
    ```

    b. For each of the previously identified nodes, do the following:

        i. Create a **debug** pod and **chroot** to the host on the storage node.

        ```
        $ oc debug node/<node name>
        ```

        <node name>
        Is the name of the node.

        ```
        $ chroot /host
        ```

        ii. Find the relevant device name based on the PVC names identified in the previous step.

        ```
        $ dmsetup ls| grep <pvc name>
        ```

        <pvc name>
        Is the name of the PVC.
        Example output:

        ```
        ocs-deviceset-xxx-xxx-xxx-xxx-block-dmcrypt (253:0)
        ```

        iii. Remove the mapped device.

        ```
        $ cryptsetup luksClose --debug --verbose <ocs-deviceset-name>
        ```

        <ocs-deviceset-name>

Is the name of the relevant device based on the PVC names identified in the previous step.

IMPORTANT

If the above command gets stuck due to insufficient privileges, run the following commands:

- Press **CTRL+Z** to exit the above command.
- Find the PID of the process which was stuck.

  ```
  $ ps -ef | grep crypt
  ```

- Terminate the process using the **kill** command.

  ```
  $ kill -9 <PID>
  ```

  <PID>
  Is the process ID.

- Verify that the device name is removed.

  ```
  $ dmsetup ls
  ```

10. Find the persistent volume (PV) that need to be deleted.

    ```
    $ oc get pv -L kubernetes.io/hostname | grep <storageclass-name> | grep Released
    ```

    Example output:

    ```
    local-pv-d6bf175b         1490Gi      RWO        Delete        Released          openshift-storage/ocs-deviceset-0-data-0-6c5pw      localblock      2d22h       compute-1
    ```

11. Delete the PV.

    ```
    $ oc delete pv <pv_name>
    ```

12. Physically add a new device to the node.

13. Track the provisioning of PVs for the devices that match the **deviceInclusionSpec**. It can take a few minutes to provision the PVs.

    ```
    $ oc -n openshift-local-storage describe localvolumeset <lvs-name>
    ```

    Example output:

    ```
    [...]
    Status:
      Conditions:
        Last Transition Time:         2020-11-17T05:03:32Z
    ```

```
    Message:                  DiskMaker: Available, LocalProvisioner: Available
    Status:                 True
    Type:                   DaemonSetsAvailable
    Last Transition Time:        2020-11-17T05:03:34Z
    Message:                  Operator reconciled successfully.
    Status:                 True
    Type:                   Available
  Observed Generation:          1
  Total Provisioned Device Count: 4
Events:
Type    Reason      Age         From                Message
----    ------      ----        ----                -------
Normal  Discovered  2m30s (x4    localvolumeset-     node.example.com -
        NewDevice   over 2m30s)  symlink-controller  found possible
                                                     matching disk,
                                                     waiting 1m to claim

Normal  FoundMatch  89s (x4      localvolumeset-     node.example.com -
        ingDisk     over 89s)    symlink-controller  symlinking matching
                                                     disk
```

Once the PV is provisioned, a new OSD pod is automatically created for the PV.

14. Delete the **ocs-osd-removal** job(s).

```
$ oc delete -n openshift-storage job ocs-osd-removal-job
```

Example output:

```
job.batch "ocs-osd-removal-job" deleted
```

NOTE

When using an external key management system (KMS) with data encryption, the old OSD encryption key can be removed from the Vault server as it is now an orphan key.

15. Scale up the **rook-ceph-operator** deployment.

```
$ oc -n openshift-storage scale deployment rook-ceph-operator --replicas=1
```

16. [Optional] Verify the Ceph cluster health and check if there are any crash reports associated with the previous failed OSD which has now been replaced.

    a. List all crash reports:

    ```
    $ oc exec -it $(oc get pod -n openshift-storage -l app=rook-ceph-operator -o name) -n openshift-storage -- ceph crash ls -c /var/lib/rook/openshift-storage/openshift-storage.config
    ```

    Sample output:

    ```
    ID                              ENTITY  NEW
    2025-09-25T07:12:45Z_1234567890   osd.5   *
    ```

```
2025-09-25T08:20:11Z_abcdef1234   mon.a   *
```

b. Filter crashes for the failed OSD from the previous step. For example, if the failed OSD is **osd.5**, run:

```
$ oc exec -it $(oc get pod -n openshift-storage -l app=rook-ceph-operator -o name) -n
openshift-storage -- ceph crash ls -c /var/lib/rook/openshift-storage/openshift-
storage.config | grep osd.5
```

Note the specif crash report from the output.

c. Archive the specific crash. Use the crash ID from the previous step:

```
$ oc exec -it $(oc get pod -n openshift-storage -l app=rook-ceph-operator -o name) -n
openshift-storage -- ceph crash archive <crash-id> -c /var/lib/rook/openshift-
storage/openshift-storage.config
```

Replace **crash-id** with the crash report.

d. Verify the crash has been archived:

```
$ oc exec -it $(oc get pod -n openshift-storage -l app=rook-ceph-operator -o name) -n
openshift-storage -- ceph crash ls -c /var/lib/rook/openshift-storage/openshift-
storage.config
```

Verification steps

1. Verify that there is a new OSD running.

```
$ oc get -n openshift-storage pods -l app=rook-ceph-osd
```

Example output:

```
rook-ceph-osd-0-5f7f4747d4-snshw    1/1     Running     0          4m47s
rook-ceph-osd-1-85d99fb95f-2svc7    1/1     Running     0          1d20h
rook-ceph-osd-2-6c66cdb977-jp542    1/1     Running     0          1d20h
```

> **IMPORTANT**
>
> If the new OSD does not show as **Running** after a few minutes, restart the **rook-ceph-operator** pod to force a reconciliation.
>
> ```
> $ oc delete pod -n openshift-storage -l app=rook-ceph-operator
> ```
>
> Example output:
>
> ```
> pod "rook-ceph-operator-6f74fb5bff-2d982" deleted
> ```

2. Verify that a new PVC is created.

```
$ oc get -n openshift-storage pvc | grep <lvs-name>
```

Example output:

```
ocs-deviceset-0-0-c2mqb   Bound    local-pv-b481410       1490Gi     RWO          localblock
5m
ocs-deviceset-1-0-959rp   Bound    local-pv-414755e0      1490Gi     RWO          localblock
1d20h
ocs-deviceset-2-0-79j94   Bound    local-pv-3e8964d3      1490Gi     RWO          localblock
1d20h
```

3. Optional: If cluster-wide encryption is enabled on the cluster, verify that the new OSDs are encrypted.

   a. Identify the nodes where the new OSD pods are running.

      ```
      $ oc get -n openshift-storage -o=custom-columns=NODE:.spec.nodeName pod/<OSD-pod-name>
      ```

      <OSD-pod-name>

      Is the name of the OSD pod.
      For example:

      ```
      $ oc get -n openshift-storage -o=custom-columns=NODE:.spec.nodeName pod/rook-ceph-osd-0-544db49d7f-qrgqm
      ```

      Example output:

      ```
      NODE
      compute-1
      ```

   b. For each of the nodes identified in the previous step, do the following:

      i. Create a debug pod and open a chroot environment for the selected host(s).

         ```
         $ oc debug node/<node name>
         ```

         <node name>

         Is the name of the node.

         ```
         $ chroot /host
         ```

      ii. Check for the **crypt** keyword beside the **ocs-deviceset** name(s).

         ```
         $ lsblk
         ```

4. Log in to OpenShift Web Console and check the OSD status on the storage dashboard.

> **NOTE**
>
> A full data recovery may take longer depending on the volume of data being recovered.

5.2. REPLACING OPERATIONAL OR FAILED STORAGE DEVICES ON IBM POWER

You can replace an object storage device (OSD) in OpenShift Data Foundation deployed using local storage devices on IBM Power.

> **NOTE**
>
> There might be a need to replace one or more underlying storage devices.

Prerequisites

- Red Hat recommends that replacement devices are configured with similar infrastructure and resources to the device being replaced.
- Ensure that the data is resilient.
  - In the OpenShift Web Console, click **Storage → Data Foundation**.
  - Click the **Storage Systems** tab, and then click **ocs-storagecluster**.
  - In the **Status card** of **Block and File** dashboard, under the **Overview** tab, verify that *Data Resiliency* has a green tick mark.

Procedure

1. Identify the OSD that needs to be replaced and the OpenShift Container Platform node that has the OSD scheduled on it.

   ```
   $ oc get -n openshift-storage pods -l app=rook-ceph-osd -o wide
   ```

   Example output:

   ```
   rook-ceph-osd-0-86bf8cdc8-4nb5t   0/1     crashLoopBackOff   0   24h   10.129.2.26
   worker-0     <none>       <none>
   rook-ceph-osd-1-7c99657cfb-jdzvz   1/1     Running   0        24h     10.128.2.46     worker-1
   <none>       <none>
   rook-ceph-osd-2-5f9f6dfb5b-2mnw9    1/1     Running   0        24h     10.131.0.33    worker-2
   <none>       <none>
   ```

   In this example, **rook-ceph-osd-0-86bf8cdc8-4nb5t** needs to be replaced and **worker-0** is the RHOCP node on which the OSD is scheduled.

   > **NOTE**
   >
   > The status of the pod is **Running** if the OSD you want to replace is healthy.

2. Scale down the OSD deployment for the OSD to be replaced.

   ```
   $ osd_id_to_remove=0
   ```

   ```
   $ oc scale -n openshift-storage deployment rook-ceph-osd-${osd_id_to_remove} --replicas=0
   ```

where, **osd_id_to_remove** is the integer in the pod name immediately after the **rook-ceph-osd** prefix. In this example, the deployment name is **rook-ceph-osd-0**.

Example output:

```
deployment.extensions/rook-ceph-osd-0 scaled
```

3. Verify that the **rook-ceph-osd** pod is terminated.

```
$ oc get -n openshift-storage pods -l ceph-osd-id=${osd_id_to_remove}
```

Example output:

```
No resources found in openshift-storage namespace.
```

> **IMPORTANT**
>
> If the **rook-ceph-osd** pod is in **terminating** state for more than a few minutes, use the **force** option to delete the pod.
>
> ```
> $ oc delete -n openshift-storage pod rook-ceph-osd-0-86bf8cdc8-4nb5t --grace-period=0 --force
> ```
>
> Example output:
>
> ```
> warning: Immediate deletion does not wait for confirmation that the running resource has been terminated. The resource may continue to run on the cluster indefinitely.
>   pod "rook-ceph-osd-0-86bf8cdc8-4nb5t" force deleted
> ```

4. Remove the old OSD from the cluster so that you can add a new OSD.

   a. Identify the **DeviceSet** associated with the OSD to be replaced.

   ```
   $ oc get -n openshift-storage -o yaml deployment rook-ceph-osd-${osd_id_to_remove} | grep ceph.rook.io/pvc
   ```

   Example output:

   ```
   ceph.rook.io/pvc: ocs-deviceset-localblock-0-data-0-64xjl
       ceph.rook.io/pvc: ocs-deviceset-localblock-0-data-0-64xjl
   ```

   In this example, the Persistent Volume Claim (PVC) name is **ocs-deviceset-localblock-0-data-0-64xjl**.

   b. Identify the Persistent Volume (PV) associated with the PVC.

   ```
   $ oc get -n openshift-storage pvc ocs-deviceset-<x>-<y>-<pvc-suffix>
   ```

   where, ***x***, ***y***, and ***pvc-suffix*** are the values in the **DeviceSet** identified in an earlier step.

   Example output:

```
NAME                    STATUS      VOLUME       CAPACITY   ACCESS MODES
STORAGECLASS   AGE
ocs-deviceset-localblock-0-data-0-64xjl   Bound    local-pv-8137c873    256Gi     RWO
localblock     24h
```

In this example, the associated PV is **local-pv-8137c873**.

c. Identify the name of the device to be replaced.

```
$ oc get pv local-pv-<pv-suffix> -o yaml | grep path
```

where, ***pv-suffix*** is the value in the PV name identified in an earlier step.

Example output:

```
path: /mnt/local-storage/localblock/vdc
```

In this example, the device name is **vdc**.

d. Identify the **prepare-pod** associated with the OSD to be replaced.

```
$ oc describe -n openshift-storage pvc ocs-deviceset-<x>-<y>-<pvc-suffix> | grep Used
```

where, ***x***, ***y***, and ***pvc-suffix*** are the values in the **DeviceSet** identified in an earlier step.

Example output:

```
Used By:    rook-ceph-osd-prepare-ocs-deviceset-localblock-0-data-0-64knzkc
```

In this example, the **prepare-pod** name is **rook-ceph-osd-prepare-ocs-deviceset-localblock-0-data-0-64knzkc**.

e. Delete any old **ocs-osd-removal** jobs.

```
$ oc delete -n openshift-storage job ocs-osd-removal-job
```

Example output:

```
job.batch "ocs-osd-removal-job" deleted
```

NOTE

The above command must reach **Completed** state before moving to the next steps. This can take more than ten minutes.

f. Change to the **openshift-storage** project.

```
$ oc project openshift-storage
```

g. Remove the old OSD from the cluster.

```
$ oc process -n openshift-storage ocs-osd-removal -p
FAILED_OSD_IDS=${osd_id_to_remove} FORCE_OSD_REMOVAL=false |oc create -n
openshift-storage -f -
```

The FORCE_OSD_REMOVAL value must be changed to “true” in clusters that only have three OSDs, or clusters with insufficient space to restore all three replicas of the data after the OSD is removed.

> **WARNING**
>
> Proceeding will permanently remove the OSD from the cluster and destroy its data. Ensure **osd_id_to_remove** is correct. Do not process multiple IDs from the same failure domain at once. Contact Red Hat Technical Support if you need assistance.

5. Verify that the OSD was removed successfully by checking the status of the **ocs-osd-removal-job** pod.
   A status of **Completed** confirms that the OSD removal job succeeded.

   ```
   $ oc get pod -l job-name=ocs-osd-removal-job -n openshift-storage
   ```

6. Ensure that the OSD removal is completed.

   ```
   $ oc logs -l job-name=ocs-osd-removal-job -n openshift-storage --tail=-1 | egrep -i 'completed
   removal'
   ```

   Example output:

   ```
   2022-05-10 06:50:04.501511 I | cephosd: completed removal of OSD 0
   ```

   > **IMPORTANT**
   >
   > If the **ocs-osd-removal-job** fails and the pod is not in the expected **Completed** state, check the pod logs for further debugging.
   >
   > For example:
   >
   > ```
   > # oc logs -l job-name=ocs-osd-removal-job -n openshift-storage --tail=-1
   > ```

7. If encryption was enabled at the time of install, remove **dm-crypt** managed **device-mapper** mapping from the OSDs that are removed from the respective OpenShift Data Foundation nodes.

   a. Get the PVC name(s) of the replaced OSD(s) from the logs of **ocs-osd-removal-job** pod.

   ```
   $ oc logs -l job-name=ocs-osd-removal-job -n openshift-storage --tail=-1  |egrep -i
   ‘pvc|deviceset’
   ```

Example output:

```
2021-05-12 14:31:34.666000 I | cephosd: removing the OSD PVC "ocs-deviceset-xxxx-xxx-xxx-xxx"
```

b. For each of the previously identified nodes, do the following:

   i. Create a **debug** pod and **chroot** to the host on the storage node.

   ```
   $ oc debug node/<node name>
   ```

   <node name>

   Is the name of the node.

   ```
   $ chroot /host
   ```

   ii. Find the relevant device name based on the PVC names identified in the previous step.

   ```
   $ dmsetup ls| grep <pvc name>
   ```

   <pvc name>

   Is the name of the PVC.
   Example output:

   ```
   ocs-deviceset-xxx-xxx-xxx-xxx-block-dmcrypt (253:0)
   ```

   iii. Remove the mapped device.

   ```
   $ cryptsetup luksClose --debug --verbose ocs-deviceset-xxx-xxx-xxx-xxx-block-dmcrypt
   ```

IMPORTANT

If the above command gets stuck due to insufficient privileges, run the following commands:

- Press **CTRL+Z** to exit the above command.
- Find the PID of the process which was stuck.

```
$ ps -ef | grep crypt
```

- Terminate the process using the **kill** command.

```
$ kill -9 <PID>
```

<PID>
Is the process ID.

- Verify that the device name is removed.

```
$ dmsetup ls
```

8. Find the PV that need to be deleted.

```
$ oc get pv -L kubernetes.io/hostname | grep localblock | grep Released
```

Example output:

```
local-pv-d6bf175b 1490Gi RWO Delete Released openshift-storage/ocs-deviceset-0-data-0-6c5pw localblock 2d22h compute-1
```

9. Delete the PV.

```
$ oc delete pv <pv-name>
```

<pv-name>
Is the name of the PV.

10. Replace the old device and use the new device to create a new OpenShift Container Platform PV.

   a. Log in to the OpenShift Container Platform node with the device to be replaced. In this example, the OpenShift Container Platform node is **worker-0**.

```
$ oc debug node/worker-0
```

Example output:

```
Starting pod/worker-0-debug ...
To use host binaries, run `chroot /host`
Pod IP: 192.168.88.21
```

```
If you don't see a command prompt, try pressing enter.
# chroot /host
```

b. Record the **/dev/disk** that is to be replaced using the device name, **vdc**, identified earlier.

```
# ls -alh /mnt/local-storage/localblock
```

Example output:

```
total 0
drwxr-xr-x. 2 root root 17 Nov  18 15:23 .
drwxr-xr-x. 3 root root 24 Nov  18 15:23 ..
lrwxrwxrwx. 1 root root  8 Nov  18 15:23 vdc -> /dev/vdc
```

c. Find the name of the **LocalVolume** CR, and remove or comment out the device **/dev/disk** that is to be replaced.

```
$ oc get -n openshift-local-storage localvolume
```

Example output:

```
NAME          AGE
localblock   25h
```

```
# oc edit -n openshift-local-storage localvolume localblock
```

Example output:

```
[...]
    storageClassDevices:
    - devicePaths:
   #   - /dev/vdc
      storageClassName: localblock
      volumeMode: Block
[...]
```

Make sure to save the changes after editing the CR.

11. Log in to the OpenShift Container Platform node with the device to be replaced and remove the old **symlink**.

```
$ oc debug node/worker-0
```

Example output:

```
Starting pod/worker-0-debug ...
To use host binaries, run `chroot /host`
Pod IP: 192.168.88.21
If you don't see a command prompt, try pressing enter.
# chroot /host
```

a. Identify the old **symlink** for the device name to be replaced. In this example, the device name is **vdc**.

```
# ls -alh /mnt/local-storage/localblock
```

Example output:

```
total 0
drwxr-xr-x. 2 root root 17 Nov  18 15:23 .
drwxr-xr-x. 3 root root 24 Nov  18 15:23 ..
lrwxrwxrwx. 1 root root  8 Nov  18 15:23 vdc -> /dev/vdc
```

b. Remove the **symlink**.

```
# rm /mnt/local-storage/localblock/vdc
```

c. Verify that the **symlink** is removed.

```
# ls -alh /mnt/local-storage/localblock
```

Example output:

```
total 0
drwxr-xr-x. 2 root root 6 Nov 18 17:11 .
drwxr-xr-x. 3 root root 24 Nov 18 15:23 ..
```

12. Replace the old device with the new device.

13. Log back into the correct OpenShift Container Platform node and identify the device name for the new drive. The device name must change unless you are resetting the same device.

```
# lsblk
```

Example output:

```
NAME                       MAJ:MIN RM   SIZE RO TYPE MOUNTPOINT
vda                        252:0    0    40G  0 disk
|-vda1                     252:1    0     4M  0 part
|-vda2                     252:2    0   384M  0 part /boot
`-vda4                     252:4    0  39.6G  0 part
  `-coreos-luks-root-nocrypt 253:0    0  39.6G  0 dm   /sysroot
vdb                        252:16   0   512B  1 disk
vdd                        252:32   0   256G  0 disk
```

In this example, the new device name is **vdd**.

14. After the new **/dev/disk** is available, you can add a new disk entry to the LocalVolume CR.

a. Edit the LocalVolume CR and add the new **/dev/disk**.
In this example, the new device is **/dev/vdd**.

```
# oc edit -n openshift-local-storage localvolume localblock
```

Example output:

```
[...]
    storageClassDevices:
    - devicePaths:
    #  - /dev/vdc
      - /dev/vdd
      storageClassName: localblock
      volumeMode: Block
[...]
```

Make sure to save the changes after editing the CR.

15. Verify that there is a new PV in **Available** state and of the correct size.

    ```
    $ oc get pv | grep 256Gi
    ```

    Example output:

    ```
    local-pv-1e31f771   256Gi   RWO    Delete  Bound  openshift-storage/ocs-deviceset-localblock-2-data-0-6xhkf   localblock    24h
    local-pv-ec7f2b80   256Gi   RWO    Delete  Bound  openshift-storage/ocs-deviceset-localblock-1-data-0-hr2fx   localblock    24h
    local-pv-8137c873   256Gi   RWO    Delete  Available
    localblock    32m
    ```

16. Create a new OSD for the new device.
    Deploy the new OSD. You need to restart the **rook-ceph-operator** to force operator reconciliation.

    a. Identify the name of the **rook-ceph-operator**.

    ```
    $ oc get -n openshift-storage pod -l app=rook-ceph-operator
    ```

    Example output:

    ```
    NAME                                  READY   STATUS    RESTARTS   AGE
    rook-ceph-operator-85f6494db4-sg62v   1/1     Running   0          1d20h
    ```

    b. Delete the **rook-ceph-operator**.

    ```
    $ oc delete -n openshift-storage pod rook-ceph-operator-85f6494db4-sg62v
    ```

    Example output:

    ```
    pod "rook-ceph-operator-85f6494db4-sg62v" deleted
    ```

    In this example, the rook-ceph-operator pod name is **rook-ceph-operator-85f6494db4-sg62v**.

    c. Verify that the **rook-ceph-operator** pod is restarted.

    ```
    $ oc get -n openshift-storage pod -l app=rook-ceph-operator
    ```

    Example output:

```
NAME                            READY   STATUS    RESTARTS   AGE
rook-ceph-operator-85f6494db4-wx9xx   1/1     Running   0          50s
```

Creation of the new OSD may take several minutes after the operator restarts.

17. Delete the **ocs-osd-removal** job(s).

```
$ oc delete -n openshift-storage job ocs-osd-removal-job
```

Example output:

```
job.batch "ocs-osd-removal-job" deleted
```

NOTE

When using an external key management system (KMS) with data encryption, the old OSD encryption key can be removed from the Vault server as it is now an orphan key.

Verfication steps

1. Verify that there is a new OSD running.

```
$ oc get -n openshift-storage pods -l app=rook-ceph-osd
```

Example output:

```
rook-ceph-osd-0-76d8fb97f9-mn8qz   1/1     Running   0          23m
rook-ceph-osd-1-7c99657cfb-jdzvz   1/1     Running   1          25h
rook-ceph-osd-2-5f9f6dfb5b-2mnw9   1/1     Running   0          25h
```

2. Verify that a new PVC is created.

```
$ oc get -n openshift-storage pvc | grep localblock
```

Example output:

```
ocs-deviceset-localblock-0-data-0-q4q6b   Bound    local-pv-8137c873       256Gi      RWO
localblock         10m
ocs-deviceset-localblock-1-data-0-hr2fx   Bound    local-pv-ec7f2b80       256Gi      RWO
localblock         1d20h
ocs-deviceset-localblock-2-data-0-6xhkf   Bound    local-pv-1e31f771       256Gi      RWO
localblock         1d20h
```

3. Optional: If cluster-wide encryption is enabled on the cluster, verify that the new OSD devices are encrypted.

   a. Identify the nodes where the new OSD pods are running.

```
$ oc get -n openshift-storage -o=custom-columns=NODE:.spec.nodeName pod/<OSD-pod-name>
```

<OSD-pod-name>

Is the name of the OSD pod.
For example:

```
$ oc get -n openshift-storage -o=custom-columns=NODE:.spec.nodeName pod/rook-ceph-osd-0-544db49d7f-qrgqm
```

Example output:

```
NODE
compute-1
```

b. For each of the previously identified nodes, do the following:

i. Create a debug pod and open a chroot environment for the selected host(s).

```
$ oc debug node/<node name>
```

<node name>

Is the name of the node.

```
$ chroot /host
```

ii. Check for the **crypt** keyword beside the **ocs-deviceset** name(s).

```
$ lsblk
```

4. Log in to OpenShift Web Console and check the status card in the OpenShift Data Foundation dashboard under Storage section.

NOTE

A full data recovery may take longer depending on the volume of data being recovered.

CHAPTER 6. REMOVING OSDS USING THE OPENSHIFT DATA FOUNDATION CLI TOOL

6.1. REMOVING OBJECT STORAGE DEVICES USING THE OPENSHIFT DATA FOUNDATION CLI TOOL

You can use the OpenShift Data Foundation command line interface (CLI) tool to automate the process of object storage device (OSD) removal. This helps to avoid the possible data loss while removing OSDs.

Prerequisites

- Download the OpenShift Data Foundation command line interface (CLI) tool. With the Data Foundation CLI tool, you can effectively manage and troubleshoot your Data Foundation environment from a terminal. You can find a compatible version and download the CLI tool from the customer portal.

Procedure

1. Identify the OSD that needs to be removed. The OSD that needs removal is in **CrashLoopBackOff** or **Error** state.

   ```
   $ oc get -n openshift-storage pods -l app=rook-ceph-osd -o wide
   ```

   Example output:

   ```
   rook-ceph-osd-0-6d77d6c7c6-m8xj6   0/1   CrashLoopBackOff   0   24h   10.129.0.16
   compute-2   <none>          <none>
   ```

2. Run the following command to remove OSD **0**:

   ```
   $ odf purge-osd 0
   ```

3. [Optional] If removal of the OSD affects placement group (PG) status, enter **yes-force-destroy-osd**.

4. Verify that the last line of the command output contains **cephosd: completed removal of OSD 0**.

5. Verify the corresponding deployment is removed:

   ```
   $ oc get deployment rook-ceph-osd-0
   ```